

TRON PROGRAMMING CONTEST 2026 SUBMISSION

SENTINEL

Adaptive Traction Control for Differential AMR using TinyML Surface Classification on
 μ T-Kernel 3.0

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Problem & System Solution

The AMR Slip Challenge

Autonomous Mobile Robots (AMRs) operating on unmapped floor surfaces (ceramic tiles, slippery vinyl, textured fabric, or heavy payload conditions) suffer severe wheel slippage.

Traditional fixed-gain PID controllers cannot adapt dynamically, leading to odometry drift, trajectory tracking failure, and actuator instability.

The SENTINEL Solution

SENTINEL embeds an on-device TinyML classifier into the 20 Hz control loop on μ T-Kernel 3.0 RTOS (STM32 Nucleo H533RE).

By extracting 84 regime-invariant physics features across current, IMU, and encoder feedback, it detects surface changes and dynamically switches PID gains within ~ 2.5 s.

Hardware & Sensor Suite



Processing Core

STM32 Nucleo H533RE (ARM Cortex-M33 with FPU) running μ T-Kernel 3.0 RTOS. Operates deterministic real-time control at 20 Hz.



Perception Array

MPU6050 6-axis IMU (accel & gyro) and dual INA219 current sensors to detect body acceleration and motor torque dynamics.

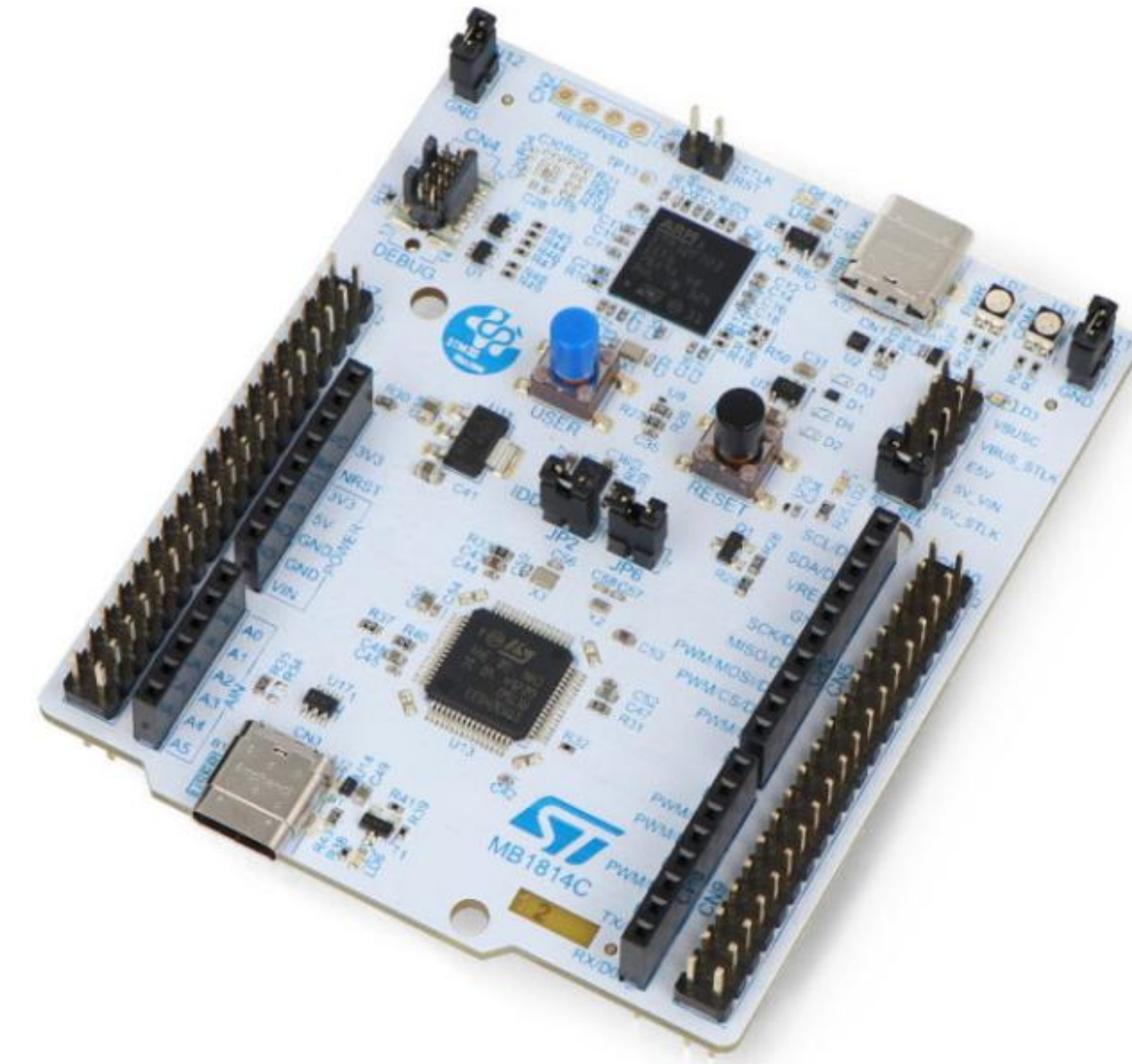


Actuation & Power

BTS7960 motor drivers driving twin PG45 DC gearmotors with optical encoders, powered by a 3S 11.1V LiPo battery.

μT-Kernel 3.0 Architecture

- ✓ **Task Hierarchy:** logTest (P5 logging), pidTask (P10 @ 20Hz), sentinelTask (P11 @ 2Hz), and comTask (P12 busy pump).
- ✓ **Low Jitter Execution:** DWT cycle counter measurements confirm control loop period jitter of only $\pm 31 \mu\text{s}$ peak-to-peak.
- ✓ **Interrupt-Driven UART:** RX ring buffer (64 bytes) prevents polling-induced byte drops during I2C sensor reads.
- ✓ **Deterministic RTOS Scheduling:** Prevents priority inversion and guarantees strictly timed wheel control.



Velocity PID & Feedforward

Velocity-Form Accumulator

Uses a velocity-form PD accumulator rather than incremental form to prevent integral windup. Pure PI accumulation is isolated from feedforward adjustments.

Rise time decreased from 550 ms to 350 ms (**-36% improvement**) while maintaining system stability.

2-Way Directional Feedforward

DC motor deadband (u_{db}) and speed constant (k_v) differ significantly between forward and reverse directions.

$$u_{ff} = \text{sgn} (\text{tgt}) \times u_{db} + \frac{k_v \times | \text{tgt} |}{KV_DEN}$$

TinyML On-Device Performance

0.54ms

Random Forest 50×10 Inference

Ultra-Light MCU Footprint

- ✓ **Total Pipeline Execution:** 3.04 ms total (2.50 ms for 84 feature extractions + 0.54 ms model execution).
- ✓ **CPU Occupancy:** Only **0.6% CPU load** against the 500 ms inference schedule budget.
- ✓ **Flash Footprint:** 141.6 KB model size compiled with per-file -Os flags via m2cgen C export.
- ✓ **RAM Footprint:** Zero dynamic heap allocation, using only 41.2 KB RAM total.

Feature Engineering Strategy



Regime Invariance

Calculates plant model residuals $(PWM - u_{ff_pred})$ and effective k_v to strip target speed variance across maneuver patterns.



Cross-Sensor Physics

Pits wheel encoder acceleration directly against MPU6050 IMU body acceleration to explicitly capture surface wheel slip.



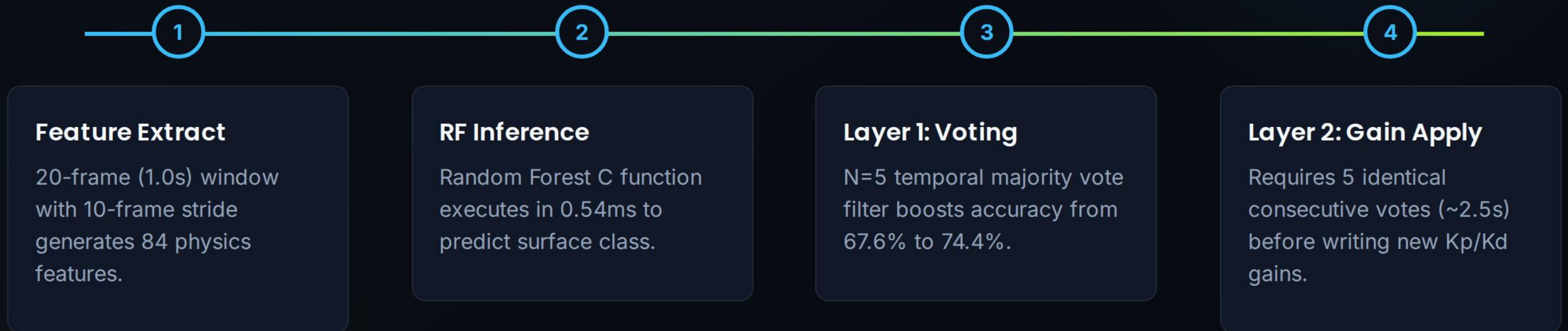
Leakage Prevention

Deliberately removed battery voltage features, eliminating session leakage and preventing a 24.5-point false confidence inflation.

Surface Classification Matrix

Class ID	Surface Material	Physical Characteristics	Scheduled Kp	Scheduled Kd	Accuracy
0: NORMAL	Clean Ceramic Tile	High traction baseline reference	70	110	80.0%
1: TEXTURED	Taut Fabric (2.56 m)	High rolling resistance	70	110	54.8%
2: BANNER	Vinyl Banner Sheet	Low friction slippery surface	70	110	59.0%
3: LOADED	Tile + 2 kg Payload	Increased mass & system inertia	85	150	77.1%

Two-Layer Hysteresis Control



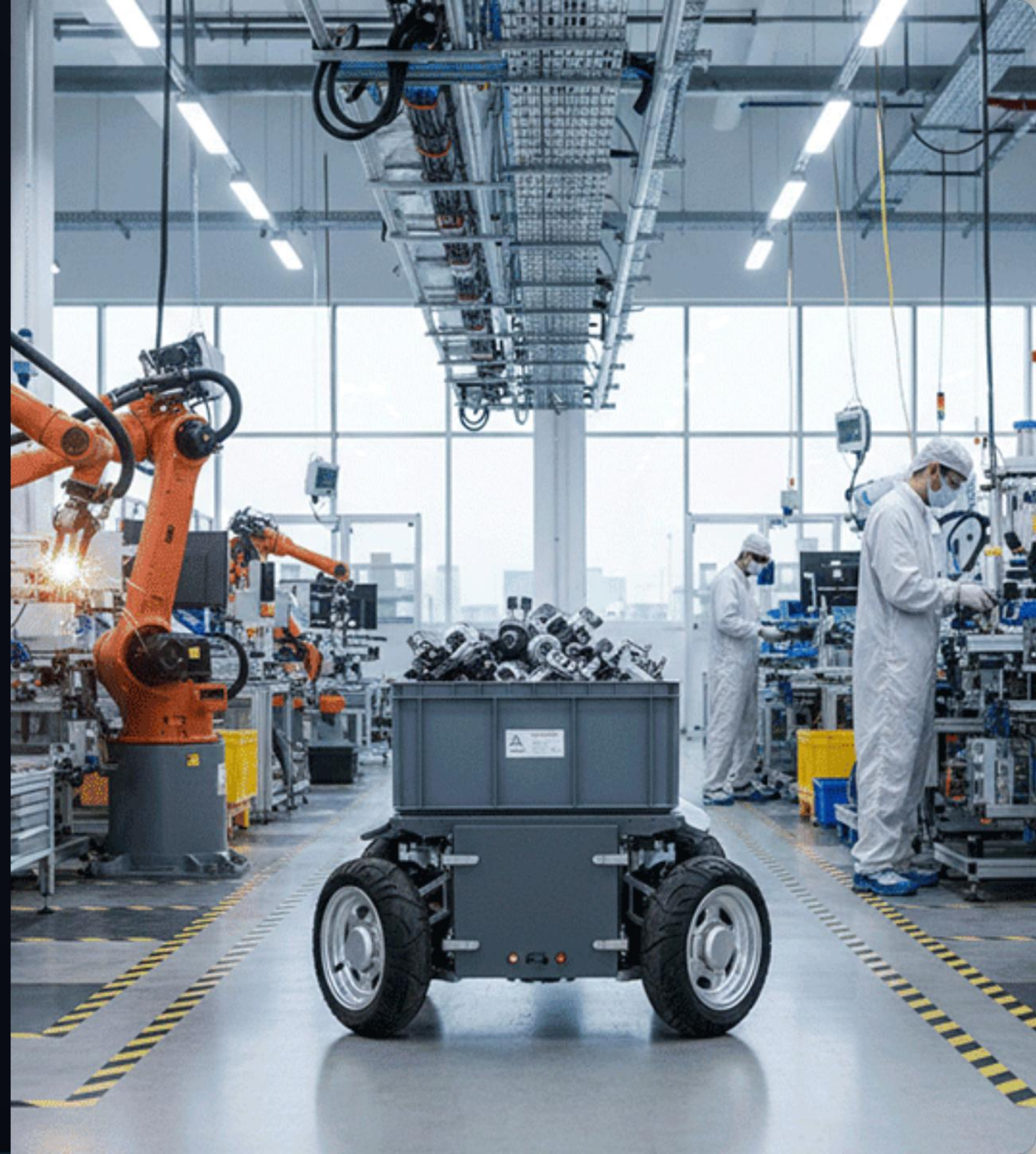
ROS2 & Dual UART Integration

Dual-Serial Line Separation

USART1 (/dev/ttySTM32): Bidirectional interface strictly dedicated to `ros2_control` (diffdrive_arduino) motor commands and encoder queries.

USART2 (/dev/ttyACM0): Isolated unidirectional telemetry line pushing CLASSIFY telemetry directly to `sentinel_classifier_node`.

This strict physical isolation guarantees zero serial bus contention or controller manager timeouts during ROS2 navigation.



Closed-Loop Step Response



📈 **Gain Scheduling Verification:** On the LOADED surface, switching to **Kp=85, Kd=150** cuts settling time by 33% (from 3300 ms down to 2200 ms) and reduces peak overshoot by 6 percentage points (from +24.7% to +18.7%).

Questions & Discussion

Thank you for watching the presentation!

 [TRON Programming Contest 2026 Submission](#)

| Image Sources



https://cdn1.botland.store/129662-large_default/stm32-nucleo-h533re-stm32h533ret6-arm-cortex-m33.jpg

Source: botland.store



<https://www.fdatobot.com/wp-content/uploads/2026/01/Autonomous-Mobile-Robot-AMR-Applications-by-Industry.png>

Source: www.fdatobot.com